Ogbonna Arinze

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SUMMARY

Staff-caliber Frontend Engineer with 8 years of professional experience architecting high-performance, scalable web applications across e-commerce-adjacent verticals including healthcare, fintech, and enterprise B2B. Deep expertise in React internals and modern Next.js (App Router, SSR, ISR, RSC), with a proven track record of designing large-scale frontend architectures, establishing engineering standards across multiple teams, and driving performance-first development. Skilled at translating complex rendering trade-offs into actionable technical strategy that delivers measurable business outcomes.

CORE COMPETENCIES

- **Frontend Architecture:** Large-scale SPA architecture, design systems, monorepos (Nx, Turborepo), SSR/ISR/SSG rendering strategies, and micro-frontends.
- **Core Technologies:** React (deep internals), Next.js (App Router, RSC, SSR, ISR), TypeScript, modern JavaScript (ESNext), and Vue.js.
- **Performance & Optimization:** Core Web Vitals, code splitting, lazy loading, CDN strategies, caching architectures, Lighthouse, bundle analysis, and React Profiler.
- **State Management & Data:** Redux, Redux Toolkit, Zustand, React Query, RxJS, and GraphQL.
- **UI Systems:** Radix UI, Tailwind CSS, Ant Design, Storybook, WCAG accessibility, and custom design systems.
- **Testing & Quality:** Playwright, Jest, Cypress, Allure Reports, ESLint, Prettier, Husky, and CI-driven QA pipelines.
- **Cloud & Infrastructure:** AWS (CloudFront, S3), Vercel, Docker, GitHub Actions, CI/CD pipelines, and Webpack/Vite/Rollup.
- **Leadership:** Cross-team architecture standardization, technical direction, mentorship, Agile/Scrum, and product-minded engineering.

WORK EXPERIENCE

Genentech (Roche) — San Francisco, CA (Remote)

Frontend Engineer — *May 2023 - Present*

- Spearheaded the redesign of the main clinical trial data product, leading the architectural migration from a **Vue.js** web application to a **React**-based Chrome extension, defining component architecture, state management patterns, and rendering strategies for the new platform.
- Architected a **reusable design system** in **React** and **Radix UI** with **Tailwind CSS**, improving UI consistency, **accessibility (WCAG)**, and developer velocity **across multiple product teams**, establishing shared component standards and usage guidelines.
- Co-developed a full-stack **Next.js** chat widget powered by a RAG system using OpenAI GPT models, LangChain, **Vercel AI SDK**, and Chroma vector database, architecting the **server-side rendering** and API layer for real-time conversational data queries.
- Optimized application performance by implementing **lazy loading, code splitting, dynamic imports**, and **CDN asset delivery**, systematically reducing load times as measured via **Lighthouse** and **Chrome DevTools** performance audits.
- Implemented **JWT-based authentication** and **role-based access control** in Next.js with **TypeScript** and **React Query**, using HTTP-only cookies for secure token handling and permission-based UI rendering across application routes.
- Introduced **Redux** for state management and **React Query** for request caching, refactoring legacy class components to functional components with hooks, streamlining data flow and improving maintainability across the application.
- Increased release reliability by integrating **Playwright end-to-end tests** into the **CI/CD pipeline** with GitHub Actions, automating reporting with **Allure Reports** on GitHub Pages and configuring Slack notifications for build quality visibility.
- Maintained and extended the existing **Vue.js** codebase during the migration period, shipping features and fixing bugs in parallel with the React rebuild to ensure zero disruption to active users.
- Mentored a team of two junior developers (interns), fostering a culture of technical excellence and code quality through structured code reviews and pairing sessions.

Descasio — Lagos, Nigeria (Hybrid)

Senior Frontend Developer — *May 2021 - Mar 2023*

- Led the design and implementation of an intuitive **React** user interface, achieving a **25% boost in user engagement** and collaborating with software architects to create a **scalable frontend architecture** that accelerated development by **30%**.
- Optimized frontend performance by profiling render bottlenecks with **React Profiler** and **Lighthouse**, reducing bundle size by **30-40%** through rollup-plugin-visualizer and **webpack-bundle-analyzer**, achieving a **35% faster load time** across devices.
- Drove a **40% improvement** in application performance by designing and optimizing **caching mechanisms** across the React application, improving speed and user experience for high-traffic pages.

- Enhanced development workflow by introducing **Jest** and **Enzyme** automated testing frameworks, leading to a **20% decrease** in bug reports and faster regression detection.
- Utilized **Storybook** for documenting React components, streamlining change management and ensuring clear communication of design patterns and component APIs across the engineering team.
- **Established engineering standards** for code quality, testing, and deployment across the team using **ESLint, Prettier, Husky, Lint-staged**, and **GitHub Actions**, reducing errors and accelerating CI/CD processes.
- Mentored junior engineers and guided frontend best practices, fostering a culture of high-quality, maintainable code and consistent design across projects.

ActivEdge Technologies Lagos, Nigeria
Senior React Developer *Oct 2019 - Sep 2021*

- Built a high-volume bank **Settlement Solution** in a team of 3 frontend engineers, handling thousands to millions of daily transactions and boosting settlement rates by **80%** through optimized **React** interfaces and data processing workflows.
- Developed critical **e-commerce** and payment modules (MasterCard, VISA, Express-cash, E-commerce) using **React** with **Redux** for state management, **RxJS** for reactive data streaming, and **Axios** for asynchronous API requests.
- Built a real-time account reconciliation solution, boosting daily reconciliation by **60%** through innovative transaction processing and data management strategies.
- Adopted the native browser **EventSource API** for real-time data streaming, replacing socket.io for faster updates, reduced latency, and a more responsive interface.
- Built dynamic data visualization dashboards using **React** and **react-chartjs**, streamlining management of user roles and permissions with **Redux Toolkit** and **Ant Design** UI.

Founders Lair Beijing, China (Remote)
Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Developed **NodeJS RESTful APIs** and translated business requirements into efficient **MongoDB** database models and schemas, collaborating closely with the business team.
- Transformed Figma designs into reusable **React** components and engineered a **CI/CD pipeline** for automated deployment to Heroku and **AWS**.
- Mentored two interns through frontend and backend projects, providing technical guidance, code reviews, and task management.

SureTrade Business Solutions Nigeria
Frontend Developer *Mar 2018 - Oct 2019*

- Designed **MongoDB** database schemas and implemented server-side logic in **NodeJS**, building and testing **REST API** endpoints for multiple client-facing applications.
- Developed **SureEdu**, a school management portal in a team of 5, handling staff payroll and attendance tracking, student records (exams, reports, attendance), and guardian communications.
- Responsible for full-stack integration, connecting the **JavaScript** frontend to **NodeJS** backend services, developing user use cases, and converting business requirements into database schemas and models.
- Devised comprehensive documentation for each application, detailing functions, capabilities, and features. Tested user-defined functions, triggers, and API endpoints to ensure reliability.

PROJECTS

OurNigeria ournigeria.ng
Senior Engineer *Feb 2026 - Present*

AI-powered Nigerian budget analysis platform. Multi-app architecture serving 37 states with 700+ budget documents.

- Architected the entire platform as an **Nx monorepo** managing **6 applications** including 3 **Next.js** frontends (citizen-facing web app, admin dashboard, marketing landing page) with **React 19**, **SSR** for public pages, and **Tailwind CSS v4**, orchestrated with pnpm workspaces.
- Built the citizen-facing **Next.js** frontend featuring 22 interactive chart types auto-generated from AI responses with **Recharts**, dark mode, URL state management, responsive mobile layouts, and shareable public conversations with **SEO optimization** (Open Graph metadata, JSON-LD schema, slug-based SSR URLs).
- Implemented **SSE streaming** for real-time chat responses with 15s keepalive heartbeats to prevent reverse proxy timeouts, delivering a smooth, performant conversational interface.
- Configured **CI/CD** with **GitHub Actions** for automated builds, **Playwright** end-to-end tests across 4 test projects, and **Docker Compose** production deployment with multi-stage Dockerfiles and health checks.

EDUCATION

University of Nigeria Nsukka, Enugu
BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*